

Empowering Self-Management of Diabetes through Long-Term Wearable Data and Seasonal Visualizations

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Abstract

Wearable devices offer unique opportunities for self-management of chronic conditions. However, long-term wearable data is often underutilized due to its complexity. In this work, we investigate opportunities around long-term wearable data from continuous glucose monitors (CGMs) used for diabetes management. Our study includes analysis of up to 7 years of wearable CGM data from 11 people with type 1 diabetes (T1D), followed by semi-structured interviews to unpack factors that affect management. Using seasonal visualizations as probes during the interviews, we found that blood glucose management is highly individual with the same factors yielding divergent outcomes across participants. We also found that seasonal visualizations of an individual's long-term data across various time horizons (e.g., months of the year, days of the week, hours of the day) can improve awareness of trends, prompt reflections on influencing factors, and inspire actionable strategies to improve outcomes. Insights from this study highlight untapped opportunities around leveraging long-term wearable data in diabetes management and inform design implications for personal informatic solutions across other continuous monitoring domains.

CCS Concepts

• **Human-centered computing** → **Empirical studies in visualization**; **Empirical studies in HCI**.

Keywords

Personal informatics, chronic disease management, type 1 diabetes, wearable devices, semi-structured interview, data visualizations

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1 INTRODUCTION

Managing a chronic condition like type 1 diabetes (T1D) is a complex and continuous process that requires making many health decisions daily. Some of these decisions include monitoring blood glucose, counting carbohydrate content of meals, calculating and administering insulin, managing physical activity and stress, and responding to high or low glucose levels [18, 19]. At the core, self-management entails balancing life demands with diabetes care while navigating the hidden burden of living with a chronic condition [21, 48]. To support the complex task of managing diabetes, wearable devices like continuous glucose monitors (CGMs) enable real-time monitoring of blood glucose values and trends to guide timely treatment decisions [16, 52] - see Fig. 1. However, CGMs generate large volumes of data over days, months, and years, thereby making it overwhelming and challenging for patients and caregivers to review prior data and extract insights. Given this, there is an immense need for data-driven solutions that facilitate discovery of actionable insights from the wealth of patient data continuously generated by wearable devices.

Previous research has shown that only a minority of CGM users revisit their retrospective data to discover patterns in their blood glucose management [4, 62]. Furthermore, standardized solutions used in clinic (e.g. the ambulatory glucose profile) only allow for short-term data review (i.e., up to 14 days) [9, 25]. As a result, long-term CGM data remains largely underutilized, despite its potential to offer valuable insights for understanding and improving diabetes management [6, 39]. Systematic reviews of technology-based tools for chronic disease management reveal several limitations, including gaps in personalization, usability and integration, and opportunities for personalized discovery for data [33, 40, 55, 59]. Individual heterogeneity in diabetes phenotypes combined with differences in daily, monthly, and yearly routines, meal habits and choices, physical activity, sleep schedules, insulin sensitivity, and more, further underscore the need for targeted solutions to support people living with diabetes [29, 48–50].

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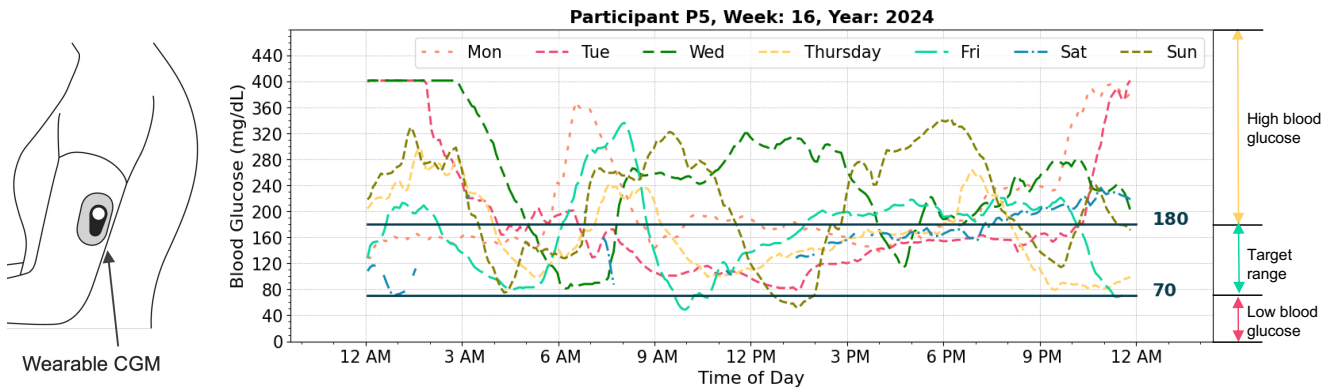


Figure 1: Wearable CGM used for daily management of diabetes (left). One week of blood glucose data from a wearable CGM for participant P5 (right).

To address these gaps, this study investigates opportunities around long-term wearable data from CGMs used for daily management of diabetes. Our work focuses on two primary research questions (RQs):

- RQ1: What unique insights can we learn from long-term wearable data as it relates to individualized factors that positively and negatively affect blood glucose management?
- RQ2: How might seasonal visualizations across various time horizons (i.e., months of the year, days of the week, and hours in the day) facilitate analysis of long-term wearable data to support self-management of diabetes?

To address the above RQs, we conducted a mixed method study including analysis of long-term CGM data (2 - 7 years of data) from 11 participants with T1D. First, we analyzed each individual's CGM data using seasonal visualizations to uncover changes in blood glucose management across various time horizons (i.e. months of the year, days of the week, and hours in the day). Following this, we used these seasonal visualizations as probes in semi-structured interviews with study participants while seeking to understand unique factors that affect diabetes management. Our findings show that blood glucose management is highly individual with the same factor yielding divergent outcomes. We also found that seasonal visualizations prompted reflection on different factors, improved awareness of the magnitude of blood glucose fluctuations, and informed actionable insights. This study contributes to the CHI community by informing design implications for personal informatic solutions that support analysis of long-term wearable data for diabetes and other continuous monitoring domains.

2 RELATED WORK

Wearable devices are increasingly being used for self-management of chronic conditions [26, 27, 31, 32, 34, 36]. In diabetes domain, CGMs and insulin pumps are clinically-validated wearable medical devices known to improve daily management and long-term outcomes [5, 14, 15]. However, several roadblocks still exist that make optimal blood glucose management elusive for many people living with diabetes. For example, there are challenges around adoption and sustained use of wearable devices, immense individual

variability in treatment response and insulin sensitivity, and difficulties managing blood glucose levels around meals, exercise, life transitions and mental health conditions [2, 24, 30, 37, 38, 45, 49, 51].

Amongst the large body of work centered around data-driven solutions in the diabetes domain, there is sufficient evidence that highlights the need for more research on self-management tools that prioritize user-centered design approaches [3, 4, 29, 46]. For example, Raj et al. developed a context-enhanced diary application to support trend identification and strategy formulation through manual logging of contextual data [47]. Research by Mamykina et al. showed that identifying cause-and-effect relationships from diabetes self-monitoring data can improve management strategies [35]. Other studies have explored various methods for enhancing diabetes management through data exploration and visualization. For example, Wong et al. introduced Blip, a tool for integrating glucose data from multiple sources to encourage user engagement and improve self-management [61]. Scimone et al. developed Marjorie, a visual analytic solution that helps uncover glucose patterns to facilitate adjustments to treatment strategies [54]. Other studies have presented computational tools, such as GlucoMine [4], cgmquantify [8], and Glucose-at-a-Glance [23] for CGM data visualization and understanding glucose fluctuations. Zhang et al. reviewed 21 diabetes management tools, including commercial applications like Tidepool [58] and Dexcom [17], and presented key design considerations related to data types, visualization views, and analytical strategies [65]. While these studies highlight the importance of CGM data visualization for self-care and clinical decision-making, there is limited work on seasonal analysis and insight discovery from years of wearable data.

Furthermore, prior studies that have investigated seasonal patterns of long-term CGM data have primarily focused on population-level trends in diabetes management. For example, Belsare et al. leveraged up to 4.5 years of wearable CGM data to understand temporal changes in glycemic trends and revealed that glycemic control often deteriorates during the holidays and colder months [6, 7]. Similarly, Oi-Yo et al. found that there are seasonal fluctuations present in blood glucose metrics with poorer glycemic control observed in winter [41]. While these studies focus on aggregate trends, there

has been limited understanding of individual variability and the personalized factors driving these seasonal changes.

3 METHOD

3.1 Study overview

Toward our goal of understanding long-term wearable data and individualized factors that affect blood glucose management, we recruited people living with T1D who were over the age of 18 years old and had been using a CGM for daily management of their diabetes for at least 2 years. Participants were recruited remotely through email solicitations to subjects from other diabetes-related projects, including the SweetGoals study [56], as well as through referrals. This study was approved by the Committee for the Protection of Human Subjects at Dartmouth College and all participants provided informed consent. As compensation, participants received a \$25 gift card for completing the study. Additionally, we deidentified all CGM data and interview records before quantitative and qualitative analysis.

Upon recruitment and providing informed consent, study participants were requested to share their CGM data with our study team through Tidepool [58] and then participated in a follow-up semi-structured interview to discuss their journey with diabetes. Before the interview, seasonal visualizations of each participant's long-term CGM data were generated in accordance with the approach presented in prior work [6]. These visualizations were then used as probes during semi-structured interviews. Each interview was conducted over Zoom, lasted around 30 minutes, and was recorded for subsequent analysis. During the interviews, participants were asked open-ended questions about their diabetes journey, including individualized factors that influence their blood glucose management. Afterwards, participants were shown several visualizations of their own CGM data and asked open-ended questions about their understanding and observations from each visual.

More specifically, participants were asked questions related to insights/trends they observed from each visual, whether the observed trends aligned with their expectation, and potential factors that contributed to the observed trends from each visualization. Finally, participants were asked if they found any of the visuals helpful for reviewing the long-term CGM data, and if so, which ones and why. Sample questions asked during the interviews include: "How would you describe your diabetes management?", "Are there certain conditions and/or situations during which your blood glucose management may be better or worse?", and "Here are some visuals of your blood glucose data, what are your observations, are there any surprises, what are your thoughts on what might have influenced the observed trends?".

3.2 Participants

Table 1 presents an overview of our study participants including a total of 11 people living with T1D - 9 females and 2 males between the age of 20 to 30 years old (mean $\pm$ std: 24.4 ± 2.9 years). Given the remote recruitment process used in this study, participants resided in various U.S. states (i.e., 10 unique states) including Utah, Oklahoma, Virginia, New York, and others. All participants used a Dexcom CGM, specifically the Dexcom G6 and G7 [17]. For insulin delivery, 8 out of 11 participants used an insulin pump or a hybrid

closed-loop system such as the Omnipod 5 [42] and Tandem t:slim [57]; however, 3 participants used multiple daily injections (MDI). Diabetes management also varied widely across study participants as evident from the hemoglobin A1C range of 6.7% - 10.2% (clinical target is $<7\%$ [1, 15]).

3.3 Data Analysis

3.3.1 Before the interview: Seasonal visualizations of blood glucose trends. Our approach to analyzing long-term CGM data was informed by prior work [6, 15, 53]. Rodbard et al. [53] highlighted several opportunities to improve the current clinically-accepted standard - ambulatory glucose profile [9] - for retrospective review of CGM data. Meanwhile, Belsare et al. [6] presented a concise method for assessing variations in blood glucose control over seasons/months of the year and days of the week. Leveraging insights from these studies, we quantified variations in diabetes management using the clinically-validated metric called time in range (TIR) [9, 15]. TIR represents the percentage of blood glucose values within the target range of 70 - 180 mg/dL as shown in Fig. 1.

In our analysis, we quantify the change in TIR compared to each individual's baseline (i.e., their own average TIR) across three distinct time horizons: months of the year, days of the week, and hours of the day. For the monthly and weekly analysis, TIR was calculated for each 24-hour day based on the available CGM data and only days with more than 70% CGM wear time were included in our analysis. Likewise, for the hourly analysis, TIR was calculated for every 1-hour window in each day that had CGM data records. Finally, to assess the change in TIR over each time horizon, we normalized by each participant's average TIR for each time horizon. Our approach leverages the method presented by Belsare et al. to elicit seasonal variations in glycemic trends [6].

Fig. 2 illustrates two examples of seasonal visualizations of long-term blood glucose management for two study participants (P1 and P11). In these visualizations, the baseline (i.e., the zero line) represents each participant's average TIR, the green bars above the zero line represent a positive change in TIR (i.e., improved blood glucose management compared to their own average), and the yellow bars represent a negative change in TIR (i.e., less than ideal blood glucose management compared to their own average). These multi-scale visualizations were designed to provide a concise and intuitive visualization of changes in blood glucose management over different time horizons for each unique participant. Across all participants in this study, the collected CGM data spanned between year 2016 and 2024, with each participant contributing between 1.8 years to 7.6 years of blood glucose data (mean $\pm$ std: 4.8 ± 1.9 years). In total, our dataset included 19,115 days of CGM data.

3.3.2 Post Interview: Thematic analysis. Upon completing all interviews, we used Otter.ai [43] to automatically transcribe the audio recordings. All automatically generated transcripts were split among three authors for review against the audio recordings to correct inaccuracies and ensure high-quality transcripts. Following this, we conducted thematic analysis using a blend of inductive and deductive approaches on the interview data [11, 12]. First, two researchers independently coded the same two transcripts (P6 and P7) to establish a preliminary set of codes and develop a shared understanding of concepts. These codes were then discussed and

Table 1: Overview of the study participants.

ID	Age	Gender	Race / Ethnicity	U.S. State	Yrs since T1D diag.	HbA1c	Insulin delivery	Dexcom CGM	Yrs of CGM data
P1	28	F	White	VA	16	6.7%	Omnipod 5	G6	4.0
P2	22	F	White	UT	9	6.9%	Omnipod 5	G6	7.6
P3	20	F	White	OK	6	7.3%	MDI	G6	2.3
P4	22	F	Hispanic	NY	14	7.6%	Pump	G6	3.1
P5	22	M	Black & White	WI	12	8.6%	MDI	G6	1.8
P6	22	M	Native Am. & White	MN	5	9.1%	Omnipod 5	G6	5.1
P7	24	F	Black	MD	21	6.8%	Tandem tslim	G7	3.4
P8	30	F	White	OR	29	7.5%	Omnipod 5	G6	6.7
P9	24	F	White	KY	23	6.9%	Omnipod 5	G7	5.6
P10	26	F	White	NY	25	7.3%	Tandem tslim	G6	7.0
P11	22	F	White	PA	9	10.2%	MDI	G7	6.0

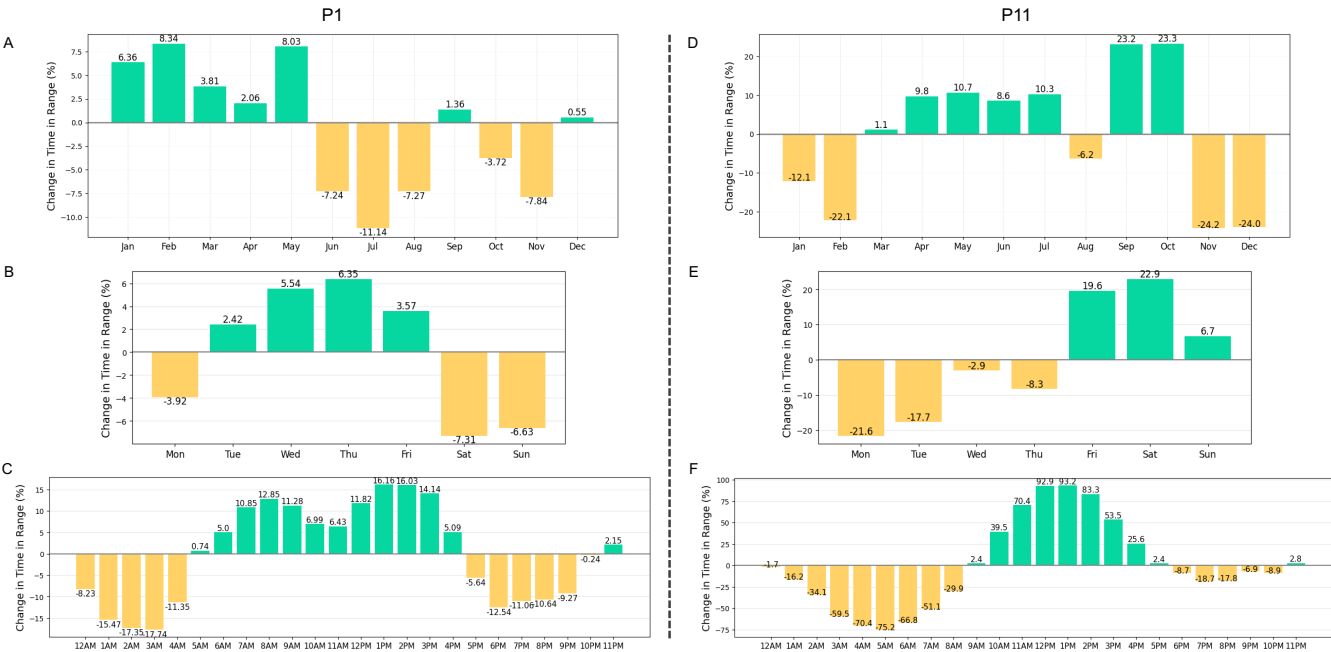


Figure 2: Seasonal visualization of blood glucose management across multiple years of data. This figure shows the change in TIR plots for P1 and P11 across different time granularities (i.e., months of the year (A and D), days of the week (B and E), and hours of the day (C and F)).

refined in collaboration with two additional researchers to establish consensus and complete the initial coding. Following this, the same two researchers divided the remaining transcripts and coded them independently. The coding process was guided by our research questions relating to seasonal and other factors that influence long-term diabetes management and opportunities around reviewing long-term CGM data with seasonal visualizations. After all transcripts were coded, we consolidated these codes into broader themes related to the aforementioned topics. These themes were discussed during a collaborative meeting with the research team and were later refined by writing analytical memos. This approach allowed us to balance the flexibility for uncovering emergent insights with

the structure needed to address specific research questions, thereby ensuring both depth and relevance in our analysis.

4 FINDINGS

In this section, we present three primary themes that address our research questions around factors that influence diabetes management (see Section 4.1) and learnings from using seasonal visualizations to assess long-term CGM data (see Sections 4.2 and 4.3).

4.1 Divergent Impact of Same Factors on Diabetes Management (RQ1)

A primary theme that emerged from our interview questions on key factors that affect blood glucose management is that the same factor can influence individuals in divergent ways (i.e. the same factor could improve management for some participants but deteriorate management for others). This theme was especially apparent when discussing blood glucose management around seasons/holidays and daily routines.

Seasons and holidays affected blood glucose management for each participant differently. Seasons that positively impacted one participant's diabetes management could negatively affect another's. For example, P4 reported worse management during winter and holidays, *"I do exercise the same, but I'm not outside as often during the winter, and I think that the rest of the year, when the days are longer, I'm outside more, so I'm more physically active, whereas during the winter, you know, maybe there's not as much movement going on. There's also a lot more holidays, so heavier carb meals, and I guess, more indulgence of like, not as healthy on foods during holidays"* (P4). Conversely, P11 experienced difficulties because they *"get dehydrated very, very easily"* in summer, which makes their *"blood sugars are wacky"* (P11). Likewise, P5 noted that mental health struggles in summer negatively impact their diabetes management, *"I think the times that I have less control over my blood sugar are the times that I'm sort of struggling mentally and I do seem to get a little bit more shut in and depressed over the summer."* (P5).

Daily routines had divergent outcomes on blood glucose levels. Regular routines, such as structured school or consistent work schedules, were generally associated with better diabetes management, while irregular or unpredictable routines negatively affected blood glucose control. For example, P3 noted that a regular routine during the school year improved their diabetes management, stating, *"it's [management] better... during my school year, just because there's more routine and more structure in it, and I eat at the same times"*. Conversely, irregular work schedules disrupted management for some participants. For example, P11 shared, *"This summer, I'm working 12 hour shifts most days, so it's [management] a little rough. I eat when I can, but they're [management is] definitely better when I have a good schedule, a tight schedule"* (P11). Similarly, P7 described how inconsistent office schedules posed challenges, *"when I'm in the office, my blood sugars are a little bit more whacky. Just because most of the time I will get lunch from somewhere around the office and I am not at home where I know what my meals are and know exactly how many carbs they [the meals] are. And any scenario, like fast food, will just, kind of like, jack up my blood sugar anyway."* When reviewing a visual of their blood glucose management they said, *"time out of range might just be that I'm not at home at that point"* (P7).

4.2 Seasonal Visualizations Prompted Reflection on Different Factors and Outcomes (RQ2a)

A second theme that emerged from our interviews when using seasonal visualizations of CGM data as probes is that different seasonal visualizations prompted reflection on different factors that affect blood glucose management. This theme was especially

apparent around monthly, daily, and hourly visualizations (see Fig. 2 A & D for examples of monthly visualizations, Fig. 2 B & E for examples of daily visualizations and Fig. 2 C & F for examples of hourly visualizations).

Monthly TIR visualizations prompted reflection on seasonal factors, such as seasonal eating habits, seasonal routine changes, and seasonal mental health issues. While reviewing monthly visualizations during the interview, P9 shared that seasonal eating patterns influence fluctuations in their blood glucose management and TIR. They shared that they consume more processed carbohydrate during the cold months and holidays, whereas more fresh produce during the warmer months. *"I'm big on Thanksgiving and Christmas. Everybody in my life kind of is. I think candy consumption and carbohydrate consumption definitely goes up that time of year. It's cold, so I feel I'm ready to eat bread and pastries. I guess it's just more readily available those months. And it affects my blood sugar more than fresh fruit and vegetables that I get in the warmer months"* (P9). Regarding seasonal routines, P4 linked lower TIR in May to transitions from college to home, stating, *"there's a decrease [in TIR] in May, because if the data is from 2021 to 2024, I've been in college all these past four years, and usually May is when the term ends. So I guess a change or transition from being in school to not being in school could possibly lead to lower time in range [blood glucose control]. I guess because of changes in routine, sometimes changes lead to more unpredictable blood glucose"* (P4). Reflecting on the monthly TIR visualizations, P3 associated better management with the motivation of a new year, *"at the start of the year, I'm feeling more motivated about it."* (P3). Meanwhile, P6 noted winter challenges tied to seasonal depression, *"January, February are around moving times as well. Also I do get seasonal bouts of just anxiety, depression and all that stuff, and that probably would not help. Then, usually by April, May I start to come back out of it"* (P6).

Daily TIR visualizations promoted reflection on routine and movement. P6 reflected on how varied routines influence blood glucose management on workdays versus rest days. They shared, *"I don't really eat as much [on workdays]. I usually work Monday, Wednesday, and Thursday. And if I don't eat as much, I don't need to bolus [insulin] as much, and on the rest days, I usually run high [i.e., have high blood glucose] either on purpose to cushion myself not go low [i.e., have low blood glucose], or I am off and I forget the bolus when I'm snacking"* (P6). Similarly, P3 described intentionally having higher blood glucose to during running activity, *"I ran a little bit higher just because I didn't want to get [blood glucose] low while I was running"* (P3).

Hourly TIR visualizations prompted reflection on sleep, food, and insulin administration. While reviewing hourly visualizations, participants discussed decisions related to daily activities. For instance, P7 recognized that the balance between unpredictable eating behaviors (e.g., night snacks) and insulin intake, *"I try to be a little bit more on top of it during dinnertime, like bolusing before [my meal]... And then sometimes I overbolus [i.e., use too much insulin]. Sometimes it's eight or nine o'clock at night, and I'll be like, Okay, I need to eat something before I go to bed"* (P7). Similarly, P11 wanted to have higher blood glucose levels overnight to avoid hypoglycemia, *"My blood sugar drops really hard at night. There are times I'm scared to go to sleep if my blood sugar is in range, so I'll eat*

a snack [and] not take insulin, my blood sugar will go up, but by the time I wake up, it's back in range" (P11).

4.3 Seasonal Visualizations Improved Awareness of the Magnitude of Blood Glucose Fluctuations and Informed Actionable Insights (RQ2b)

A third theme that emerged when using seasonal visualizations of CGM data as probes during the interviews is that seasonal visualizations improved awareness of blood glucose fluctuations and supported the discovery of actionable insights.

Seasonal visualizations improved awareness of hourly, daily and monthly fluctuations of blood glucose. For example, P4 described they were less aware of the weekly patterns before seeing the visualization, *"it was helpful for me to see this, to kind of just see the patterns I didn't maybe really realize before"* (P4). Even when participants were aware of the general trend, seasonal visualizations helped them understand the magnitude of changes. For instance, P2 was surprised when they observed how different the TIR (i.e., blood glucose management) was during the winter months. *"November and December, I have no idea, which is weird, because that's kind of a huge jump, like significantly lower [blood glucose control], but I don't know if I have a good explanation for why that is"* (P2). For some participants, analysis of long-term CGM data with seasonal visualizations provided different insights beyond what is typically observed during clinic visits. P8 shared, *"I think, from a troubleshooting perspective for me, when I go into the endocrinologist they're looking at the last six weeks, ... but what does that look like over a bigger scale... I think that's probably the most useful. Now I can say, Okay, what do I need to do during ski season differently than I need to do in the summer? Or, what is the difference?"* (P8).

Increased awareness of both suboptimal and optimal diabetes management habits and trends resulted in actionable insights. Periods of suboptimal blood glucose management informed actionable insights on how to make treatment adjustments. P1 stated, *"I think it's really interesting, and I thought this was super insightful to my own patterns that now I could say, all right, it's the summer now. Maybe let me concentrate on what's going on in July and August, and make sure that I'm trying to stay more in range. Or, once the holiday season comes around, let's be more mindful of what's going on. That is something that I could take home and focus on. You know that is a way that [Dexcom] Clarity doesn't really present to you"* (P1). P11, who intentionally erred towards higher blood glucose levels to avoid hypoglycemia overnight, realized the extent of the issue from hourly visualizations showing lower TIR at night. This observation helped them identify actionable changes to their pre-bed eating habits. *"It really puts stuff into perspective for me, kind of how often I'm letting myself go that high. It's seeing that my blood sugar is typically peaking at four or five AM. So honestly, I'm probably going to start eating a little less than what I typically would eat, for my snack at night. Or maybe I'll start taking some insulin for it to see if I can bring that down"* (P11). Moreover, positive trends inspired participants to replicate success. Looking at the hourly visualizations, P3 reflected on how routines helped, *"I would say that's my school day, right there that's the green, I thrive with routines"*. Later they mentioned, *"okay, I need to have a really good*

routine, because that really helps my body" (P3). Even though P5 was frustrated over their recent blood glucose management, they felt reassured by past successes, *"throughout the last few months, I've been like, Man, my blood sugar is all over the place all the time. I don't understand it. I'm getting frustrated. I know I have had some good months before too. So I don't understand what's going on. It [seasonal visualizations] almost in a way confirms that I can do this, I've been in range before, and I'm going to do it again, and I'll be alright"* (P5).

5 DISCUSSION

This study sought to investigate opportunities around long-term wearable data from CGMs used for daily management of diabetes. In particular, our study focused on understanding unique insights that can be learned from long-term wearable data in relation to individualized factors affecting blood glucose management. We also explored opportunities for seasonal visualizations to facilitate the analysis of long-term wearable data. Through quantitative and qualitative analysis, we found that diabetes management is highly individual and the same factor can have divergent impact on outcomes. In addition, we found that seasonal visualizations offered unique insights. For example, monthly visualizations facilitated reflection on factors such as seasons and holidays, weekly visualizations captured routine variability, and hourly visualizations facilitated reflection on daily behaviors related to sleep, eating habits, and insulin use. Seasonal visualizations also provided perspectives beyond typical discussions during clinical visits, such as long-term trends across seasons and the impact of irregular routines.

While reviewing the seasonal visualizations during interviews, participants recognized the value of transforming long-term data into personalized visual representations. This approach enabled them to identify opportunities to improve blood glucose management through actions such as adjusting daily routines, incorporating exercising, reducing pre-bedtime snacks, or preparing for challenging periods like holidays or seasonal transitions. To further improve personal discovery, interactive solutions can support agency by enabling users to select specific data types for analysis and/or the time scale, and include goal-setting modules to help users plan and track behavior changes. For example, in the event that a solution detects worse than average blood glucose management during winter months, it could suggest strategies such as adjusting insulin doses or planning physical activities to mitigate seasonal effects. Similarly, in the event that a user's data show consistent high blood glucose after meals, the solution could provide tailored recommendations to improve the timing of bolus insulin doses or portion adjustment. Such features can improve user engagement, promote proactive self-management behaviors, and improve long-term health outcomes. While the potential for such a solution seems promising, practical roadblocks such as limited interoperability across wearable and other devices can make realization of a comprehensive solution challenging. Furthermore, it may be difficult for a user to concretely pinpoint reasons for better or worse blood glucose management during certain periods, thereby limiting the remedial action that can be taken following a recommendation.

Beyond the diabetes domain, seasonal visualizations can be relevant in other health and continuous monitoring domains. For

instance, understanding seasonal changes in sleep, psychiatric disorders, and mental health sensitivity are areas that all affect quality of life and well-being [10, 20, 44, 60, 63, 64]. In the clinical domain, long-term ECG monitoring can support cardiovascular health by detecting postoperative atrial fibrillation and ensuring reliable signal acquisition [13, 22]. Analyzing seizure cycles can reveal insights about circadian and multi-day rhythms, aiding personalized seizure prediction and management [28]. As a result, future solutions have the potential to improve clinical and self-management strategies, enhance patient engagement, and improve long-term health outcomes across diverse conditions.

6 Limitations and Future Work

Albeit unique strengths and insights gained from this study, there are important limitations that should be acknowledged. First, this study comprised of a relatively small size including only 11 participants, majority of whom are female. As expected, more insights can be generated from a larger and more gender-balanced population. Second, all participants used Dexcom CGMs. This limitation is an artifact of our CGM data collection process through Tidepool as data sharing for Dexcom users only requires user authorization. In contrast, prospective participants that use other non-Dexcom CGM (e.g. Medtronic or Abbott CGMs) require a multi-step process that includes physically connecting their CGM device to a computer to upload their data before they could be a part of this study. This cumbersome process made it challenging for non-Dexcom users to join our study. Third, there are limitations to our approach to collecting long-term CGM data in that periods when participants may have used a non-Dexcom CGM or other non-connected glucose monitoring devices showed up as missing data in our retrieved dataset. Lastly, all seasonal visualizations analyzed and presented in this study were static. As a result, our study participants were unable to interact with visualizations of their long-term CGM data and control/select important features relevant to the visualizations such as the duration of data plotted for review or the metric used for summarizing CGM data.

To address these limitations, future work should investigate opportunities around interactive solutions that support reflection and insight discovery from long-term wearable data in diabetes and other continuous monitoring domains. Such solutions can allow users to select and adjust various parameters such as data duration (e.g. specific months or weeks for review), summary metrics (e.g. TIR or mean blood glucose) and integrate additional data sources that support reflection in the context of other factors such as physical activity, sleep patterns and more. Such flexibility can provide a more holistic view of factors that affect management outcomes and further empower users to align data exploration with their personal management goals.

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